However, a ΔY transformation might create series or parallel edges, which can then be SP-reduced.

We note here a simple lemma, characterizing when connectivity is preserved under Y-to-Δ operations.

Lemma 4.2.

(i) *Let G be a 2-connected graph, and let $\{e, f, g\}$ be the edges at a vertex $\boldsymbol{v}$ of degree 3 in G.*

If none of its edges are parallel (i.e., if $\boldsymbol{v}$ has three different neighbors), then the result of a Y-to-Δ operation is again 2-connected.

(ii) *Let G be a 3-connected graph (in particular, there are no parallel edges; all vertex degrees are at least 3) that is not K_4. Let $\{e, f, g\}$ be the edges at a vertex $\boldsymbol{v}$ in G of degree 3.*

If we perform a Y-to-Δ operation on this 3-star, and then delete all parallel edges created by this (i.e., all edges that originally connected neighbors of $\boldsymbol{v}$), then the resulting graph is again 3-connected.

Proof. This follows directly from the definitions: for this consider a Y-to-Δ operation $G \longrightarrow G'$. Then for any set of one or two separating vertices in G', the same set is separating in G as well. The only problem is that a Y-to-Δ operation can create parallel edges; if we delete them, the operation preserves 3-connectedness. □

The nice thing now is that, because of duality, we immediately get a dual statement, Lemma 4.2*, about connectivity after a Δ-to-Y transformation. For this we use that under duality, we have

$$\begin{array}{rcl} \text{embedded planar graph } G & \longleftrightarrow & \text{dual graph } G^*, \\ \text{contracting series edges} & \longleftrightarrow & \text{deleting parallel edges}, \\ k\text{-connected} & \longleftrightarrow & k\text{-connected}, \\ \text{nonseparating triangle} & \longleftrightarrow & \text{3-star}, \\ \Delta\text{-to-}Y \text{ transformation} & \longleftrightarrow & Y\text{-to-}\Delta \text{ transformation}. \end{array}$$

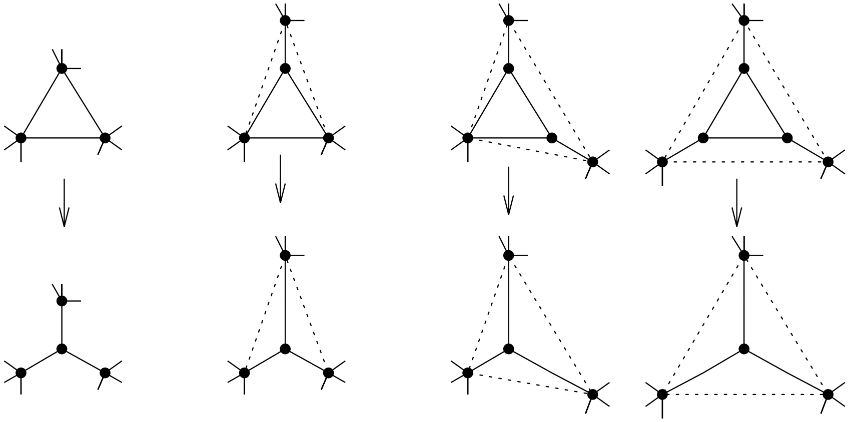
This duality can be carried into our reduction for polytopes, because the graph of a 3-polytope is exactly the dual graph of the polar polytope:

$$G(P)^* \;=\; G(P^\Delta).$$

4.2 Simple ΔY Transformations Preserve Realizability

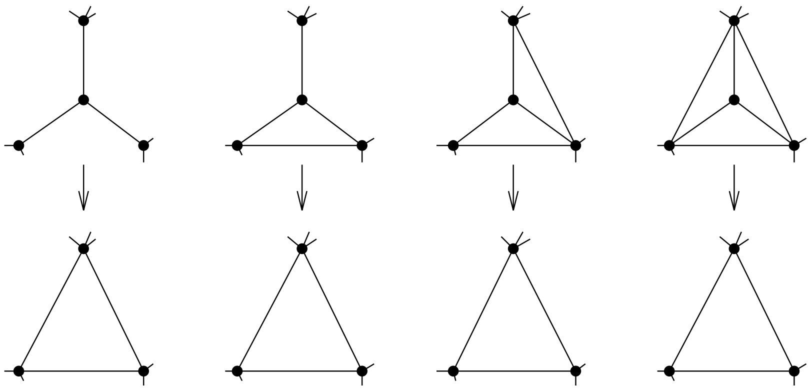
By a *simple ΔY reduction* we mean any ΔY operation followed immediately by all the SP reductions that are then possible. By Lemma 4.2 and its dual, these reductions preserve 3-connectedness, if applied to any 3-connected graph other than K_4.

We get four different types of simple Δ-to-Y reductions: for this we consider a triangle and distinguish whether it has zero, one, two, or three vertices of degree 3.



Here the dotted lines in our sketch denote edges that may or may not be present, and are not affected by the simple Δ-to-Y reduction.

Similarly, we get four types of simple Y-to-Δ transformations when we consider a vertex $\boldsymbol{v}$ of degree 3 and distinguish how many of its neighbors are already connected:



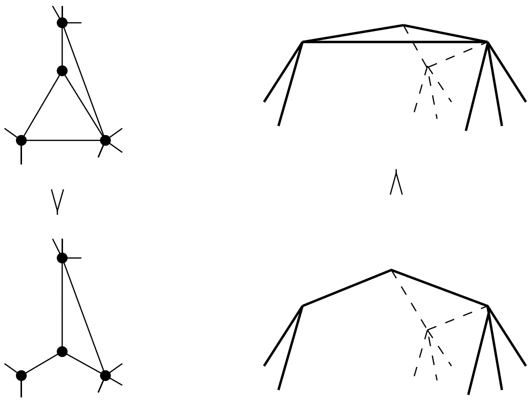
These four transformations are exactly the "polar operations" (operations in the dual graph) for the simple Δ-to-Y reductions.

Lemma 4.3. *Let G be a 3-connected planar graph, and let the graph G' be derived from G by a simple ΔY transformation.*

If G' is the graph of a 3-polytope, then so is G.

Proof. By duality, respectively polarity, we have to treat only the four types of Δ-to-Y transformations. For these, the transformation from P' to P just corresponds to "cutting off a vertex" by some suitable plane.

(To visualize this, consider our sketch of the four types of simple Δ-to-Y transformations, interpreting them as pictures of 3-polytopes.)



∎

4.3 Planar Graphs are ΔY Reducible

In this section we show that every 3-connected planar graph (with $n \geq 4$ edges) can be reduced to K_4 by a sequence of simple ΔY transformations. This is a special case of a much more powerful theorem (for 2-connected planar graphs plus a "return edge") that was first established by Epifanov [198] and has a clever and simple proof by Truemper [546]. In this section, we follow his expositions in [546] and [547, Sect. 4.3].

For a while, we will only require that the graphs considered are 2-connected; in particular, we admit ΔY operations if they keep our graphs 2-connected.

In the simplicity of Truemper's approach to Epifanov's theorem, the reader should appreciate the "power of a normal form theorem" (in this case: the embedding of a planar graph as a minor of a grid graph, which is also at the heart of Robertson & Seymour's work [464] on graph minors).

Here come three lemmas and a corollary, which together prove everything. [*Working through this, you can practice "three levels of reading": first read only the lemmas, and try to understand what they mean and how much is trivial. Second glance over the proofs, and try to see whether you know how to do them yourself. If you know, just do it. If you don't, try to find counterexamples. In the third step, work your way through the proofs until you are confident that you found all the errors I made and the shortcuts I missed. Tell me about them.*]